

Example

$arr = [1, 6, 11, 5]$

Step 1: Find total sum (range)

$range = 1 + 6 + 11 + 5 = 23$

$n = 4$

Step 2: DP Table Meaning

$t[i][j] = \text{true} \rightarrow$ we can form sum j using first i elements

Step 3: Initialization

$t[i][0] = \text{true}$ (sum 0 always possible)

Step 4: Fill DP Table

We track only important values (not full table to keep it readable).

Using first element (1)

Possible sums:

$\{0, 1\}$

Using first 2 elements (1, 6)

Possible

sums:

{0, 1, 6, 7}

Using first 3 elements (1, 6, 11)

Possible sums:

{0, 1, 6, 7, 11, 12, 17, 18}

Using all 4 elements (1, 6, 11, 5)

Add 5 to previous sums:

{0, 1, 5, 6, 7, 11, 12, 16, 17, 18, 22, 23}

Step 5: Find minimum difference

We check from:

$$\text{range}/2 = 23/2 = 11$$

Now go downward:

Check $j = 11 \rightarrow \text{YES}$

(present)

So:

$$\begin{aligned}\text{min difference} &= \text{range} - 2*j \\ &= 23 - 2*11 \\ &= 23 - 22 \\ &= 1\end{aligned}$$

Final Answer

Minimum Subset Sum Difference = 1